

YUVARAJ R

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SUMMARY

Computer Science and Engineering undergraduate with two software internships, building AI-driven backend systems across LLM applications, computer vision, and geospatial services. Ships end-to-end products with Python, TypeScript, FastAPI, Next.js, PostgreSQL, and Docker. Daily Linux (NixOS) user with self-hosted infrastructure experience. Seeking a software engineering or AI/ML internship.

EXPERIENCE

Synergech Technologies India Pvt Ltd — Software Engineering Intern | On-site, Chennai | May 2026 – Jun 2026

- Built **IntelliRender**, a streaming AI chat application that renders interactive visual widgets inline instead of plain text, on Next.js 14 App Router with TypeScript.
- Designed a type-safe widget protocol using Zod schemas and TypeScript discriminated unions, so the LLM returns validated structured JSON that a registry-pattern renderer maps to React components without touching the chat pipeline.
- Implemented SSE streaming via the Vercel AI SDK plus 3D physics (react-three-fiber), Recharts dashboards, and drag-and-drop Kanban widgets; added NextAuth, a Drizzle ORM data layer, and Docker Compose deployment.

Geospecific Solutions — Software Engineering Intern | Remote | May 2025 – Jul 2025

- Built **Wayline**, a self-hosted routing and geocoding platform on OSRM and PostgreSQL, removing dependence on paid third-party mapping APIs.
- Designed an API gateway architecture fronting the routing services and containerised the stack with Docker for repeatable, scalable deployment.

EDUCATION

Vellore Institute of Technology, Chennai — B.Tech, Computer Science and Engineering | 2024 – 2028
Chennai, India | CGPA: 7.86 / 10 (through 4th semester)

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, Dart, Solidity

AI & Machine Learning: YOLOv8, Computer Vision, Scikit-learn, NLP, OpenAI Whisper, Llama 3 (Groq), Ollama, Vercel AI SDK, MCP, MATLAB Simulink

Backend: FastAPI, Flask, Node.js, Express, REST API Design, Pydantic v2, Celery, NextAuth, JWT

Databases: PostgreSQL, PostGIS, Neo4j, Redis, Supabase, Drizzle ORM

Frontend: Next.js 14 (App Router), React, React Native (Expo), Flutter, Tailwind CSS, shadcn/ui, Three.js (R3F), Recharts

DevOps & Systems: Docker, Docker Compose, Git, Linux (NixOS), OSRM, pytest, ruff, Self-hosted Infrastructure

Embedded & Blockchain: Raspberry Pi, ESP8266, Arduino, KiCad, Smart Contracts, Polygon, IPFS

PROJECTS

DevOrchestrator — AI-Native SDLC Orchestrator | Python 3.12, Typer, Pydantic v2, Supabase, FastMCP, tmux | FRONTIER 2026 Hackathon

- Team of 4 built a CLI automating the develop-to-PR loop: two agent sessions research the codebase and implement the plan, quality gates auto-repair failures, and a PR opens with an AI-written description.
- Owned the integrations lane — GitHub Projects v2 board and git adapters, task selection, and demo tooling — against a frozen contracts layer that let four developers work in parallel without merge conflicts.
- Shipped with 278 passing tests and a Supabase-backed shared memory exposed over MCP, so any agent can query who is working in which module.

MechSight AI — Industrial Defect Inspection System | Python, YOLOv8, Flask, Raspberry Pi, Docker

- Engineered an end-to-end quality inspection pipeline using YOLOv8 to detect manufacturing defects in automotive washers and bushes, replacing manual visual checks.
- Developed a centralised Flask inference server exposing REST endpoints for real-time detection and automated defect logging, paired with a Raspberry Pi edge client for physical actuation and operator feedback.

BhoomiNode — Blockchain AgriTech Platform | FastAPI, React Native, Groq (Llama 3), PostGIS, Neo4j, Solidity | AGRHINOVA Hackathon, Apr 2026

- Built a voice-first platform letting farmers register land, receive an AI-scored Kisan Passport, and file verified crop insurance claims, delivered end to end in a 24-hour hackathon.
- Integrated Groq Whisper for regional-language speech-to-text and Llama 3 for intent extraction, with NDVI crop-health scoring from Microsoft Planetary Computer satellite data.
- Deployed land-ownership NFTs and automated insurance verdicts via Solidity smart contracts on the Polygon Amoy testnet, backed by a PostGIS and Neo4j data layer.

RESEARCH

Electrical Fault Detection and Classification — Modelled power-system fault scenarios in MATLAB Simulink to generate a labelled dataset, then benchmarked eight machine learning models to identify the strongest classifier.

ML-Based Drift Correction for CGM Sensors — Researched ML techniques for correcting sensor drift in continuous glucose monitors to improve long-term accuracy.

ACHIEVEMENTS & ACTIVITIES

- Rajya Puraskar Awardee**, Bharat Scouts & Guides — state-level award for sustained leadership and service. **Hackathons:** FRONTIER 2026; AGRHINOVA, CeAT-RC, VIT Chennai (Apr 2026).